PORTFOLIO PROJECT | PHASE A RESULTS

Natural Frequency of an Offshore Monopile Model: Validation, Tip Mass and Soil Stiffness

Project	Vibration and fatigue estimation of a bare offshore monopile before commissioning (self-study)
This phase	Build and validate a beam model in Python; study the effect of tip mass and soil stiffness on the first natural frequency
Tools	Python 3 (NumPy, SciPy, Matplotlib); open-source only
Status	Phase A core completed. All geometry and soil values are illustrative, not site data.

1. Objective

Before estimating fatigue, the dynamic behaviour of the structure must be understood. This phase answers three questions: (1) can a self-written finite element model reproduce known analytical results, (2) how does a heavy nacelle at the top change the natural frequency, and (3) how sensitive is the frequency to soil stiffness at the base?

2. Model and inputs

Parameter	Value	Parameter	Value
Young's modulus E	210 GPa	Outer diameter D	6.0 m
Steel density	7850 kg/m^3	Wall thickness t	0.07 m
Length L	60 m (varied later)	Second moment of area I	5.73 m^4
Cross-section area A	1.30 m^2	Mass per metre	10237 kg/m

The structure is a uniform steel tube modelled with Euler-Bernoulli beam elements (2 nodes, deflection and rotation per node) and a consistent mass matrix. Free vibration is solved as the generalised eigenproblem K x = lambda M x, with lambda = omega^2 and f = omega / (2 pi). The base is held by a lateral spring k_x and a rotational spring k_r; very large values reproduce a fixed base. The nacelle and rotor are represented by a lumped mass at the top node.

3. Validation against theory

With the tip mass removed and a fixed base, the model should match the closed-form cantilever solution f = (beta L)^2 / (2 pi) x sqrt(EI / (m L^4)), with beta L = 1.8751, 4.6941 and 7.8548 for the first three modes.

Case	Mode 1 (Hz)	Mode 2 (Hz)	Mode 3 (Hz)
Analytical (fixed-free)	1.686	10.564	29.580
FE, 1 element	1.694	-	-
FE, 2 elements	1.687	10.654	36.033
FE, 5 elements	1.686	10.569	29.686
FE, 10 elements	1.686	10.565	29.587
FE, 20 elements	1.686	10.564	29.580

The first mode is accurate with only 2 elements; higher modes need more. With 20 elements all three modes agree with the analytical values to three decimals, so the model is considered validated.

4. Effect of boundary conditions (analytical)

Only the factor beta L changes with the support condition. Stiffer restraint gives a higher frequency (same tube, L = 60 m).

Support	beta L (mode 1)	f1 (Hz)
Fixed-free (cantilever)	1.8751	1.686
Pinned-pinned	3.1416	4.732
Fixed-pinned	3.9266	7.392
Fixed-fixed	4.7300	10.727

5. Effect of length

Frequency scales approximately with $1 / L^2$. Increasing L from 60 m to 100 m reduces f1 from 1.686 Hz to 0.607 Hz, matching the prediction $1.686 \times (60/100)^2 = 0.607$ Hz.

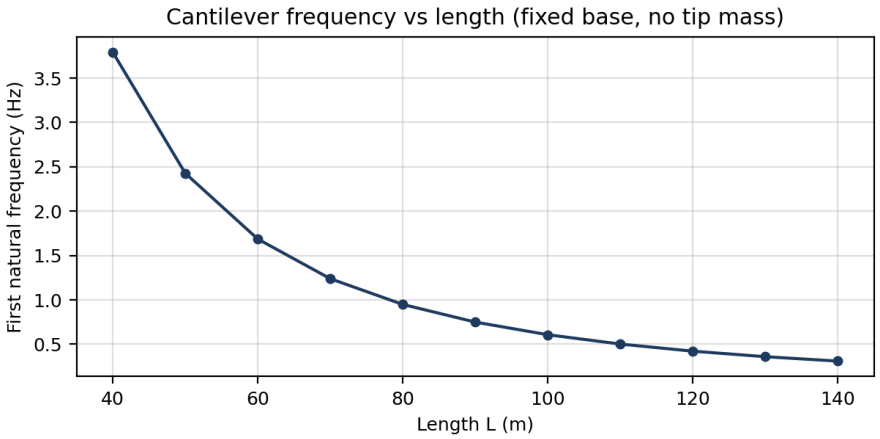


Figure 1: First natural frequency of the fixed-base cantilever versus length.

6. Effect of tip mass (nacelle and rotor)

Adding mass at the top lowers the frequency. FE results for the fixed base:

Tip mass (t)	0	200	400	800
f1 (Hz)	1.686	1.106	0.881	0.669
Reduction vs. no mass	-	34 %	48 %	60 %

The FE values agree with the engineering approximation f1 / sqrt(1 + M_{tip} / (0.24 m L)) within about 1 %.

7. Effect of soil stiffness

The rotational base stiffness k_r was varied from 10^9 to 10^{13} Nm/rad with a 400 t tip mass.

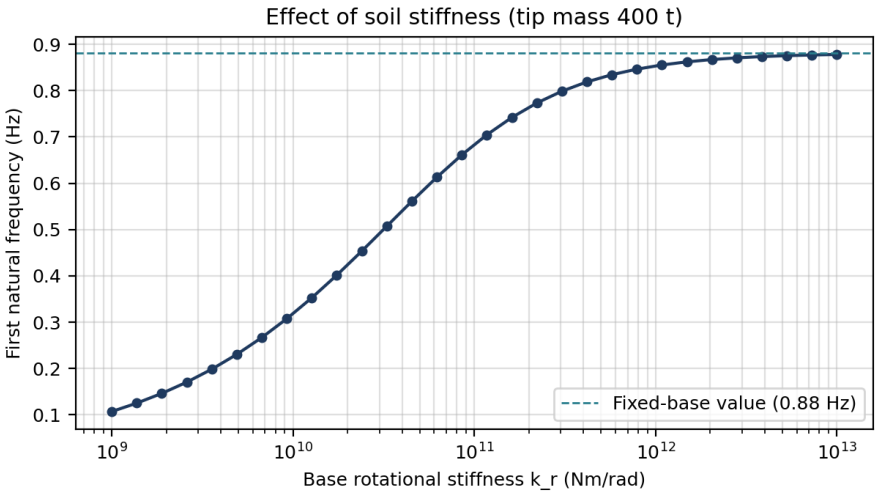


Figure 2: First natural frequency versus base rotational stiffness.

k_r (Nm/rad)	1e13	1e12	1e11	1e10	1e9
f1 (Hz)	0.878	0.853	0.684	0.318	0.107

8. Interpretation

- **Stiff soil (above about 10^{12} Nm/rad):** the base behaves as fixed and the frequency levels out near 0.88 Hz. Further stiffening changes almost nothing.
- **Intermediate soil (about 10^{10} to 10^{12} Nm/rad):** the frequency is very sensitive to soil stiffness. Uncertainty in soil properties directly becomes uncertainty in frequency here.
- **Soft soil (below about 10^{10} Nm/rad):** the soil controls the response and the frequency drops towards 0.1 Hz.
- The soil spring and beam act like springs in series: the softer one dominates.
- Wave energy lies at low frequency (roughly 0.05 to 0.2 Hz), so a structure whose frequency is pushed down by soft soil or added mass moves closer to the excitation range. This links directly to the fatigue question of later phases.

9. Limitations

- Geometry, tip mass and spring values are illustrative. The tube is shorter and stiffer than a real monopile and tower, so frequencies are higher than the 0.25 to 0.35 Hz typical of real turbines.
- The soil is represented by two springs at the base; a real pile interacts with soil along its embedded length (distributed p-y springs).
- Linear behaviour and small strains only; no soil degradation, damping or hydrodynamic added mass.
- No loading yet: this phase studies the structure only.

10. Next steps

- Build a realistic monopile: embedded length, water depth, transition piece and tower with varying section.
- Replace base springs with distributed Winkler springs (stiffness increasing with depth).
- Compare four installation stages: bare pile, with transition piece, with tower, with nacelle.
- Phase B: JONSWAP wave spectrum, Morison loads and a VIV check; then response, rainflow counting and fatigue damage.